



5907 - Characterizing the $z > 1$ satellite population with public deep-field JWST surveys

Cycle: 3, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

The abundances and radial distributions of satellite galaxies can be used to constrain both our dark matter models and the galaxy--halo connection. While existing observations of the satellite population at $z < 0.2$ have proven a rich dataset against which to test our models, to date it has been difficult or impossible to observationally measure the satellite population at cosmological distances. The advent of JWST has opened the door to measurements of faint satellites ($< 1/10$ the mass of their host) at $z > 1$ for the first time, showing that these tiny galaxies are easily resolved in deep-field JWST imaging. Here, we propose to use existing public data to perform a complete selection of satellites around a set of both star-forming and quiescent hosts at $1 < z < 3$ and make the first measurements of the number density, radial distribution, and galactic conformity of low-mass satellites at $z > 1$. We will pair these first-of-their-kind observations with equally novel theoretical predictions for the high-redshift satellite population which take advantage of recently developed halo finder tools and novel star-tagging methods that can track simulated subhalos down to small radii through

intense tidal disruption. We will predict the observed satellite number counts, radial distribution, and conformity of satellites in our observed mass and redshift range and assess whether model predictions agree with the data.

OBSERVING DESCRIPTION

We propose an systematic archival search for faint satellite galaxies at $1 < z < 3$ in public deep-field JWST imaging from JADES, CEERS, and PRIMER. These tiny companions were invisible in previous ground- or space-based imaging, but have been shown to be detectable down to stellar masses of $\sim 10^7 M_\odot$ with JWST imaging. As described in our analysis plan, we will (1) measure the number density, radial distribution, and conformity of satellites around massive hosts from public JWST data, and (2) carefully compare these observations with predictions from simulations. We expect to publish at least two papers from this analysis, one presenting the observations and one with a detailed model comparison.